



# Introduction to InterSystems IRIS Business Intelligence

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*Introduction to InterSystems IRIS Business Intelligence*

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# 1

## Introduction to Business Intelligence

This page introduces InterSystems IRIS Business Intelligence, which enables you to embed business intelligence (BI) into your applications.

**Note:** You can use Business Intelligence only in a specifically configured web application. See [Setting Up the Web Application](#). Also, be sure to consult *InterSystems Supported Platforms* for information on system requirements.

### 1.1 Purpose

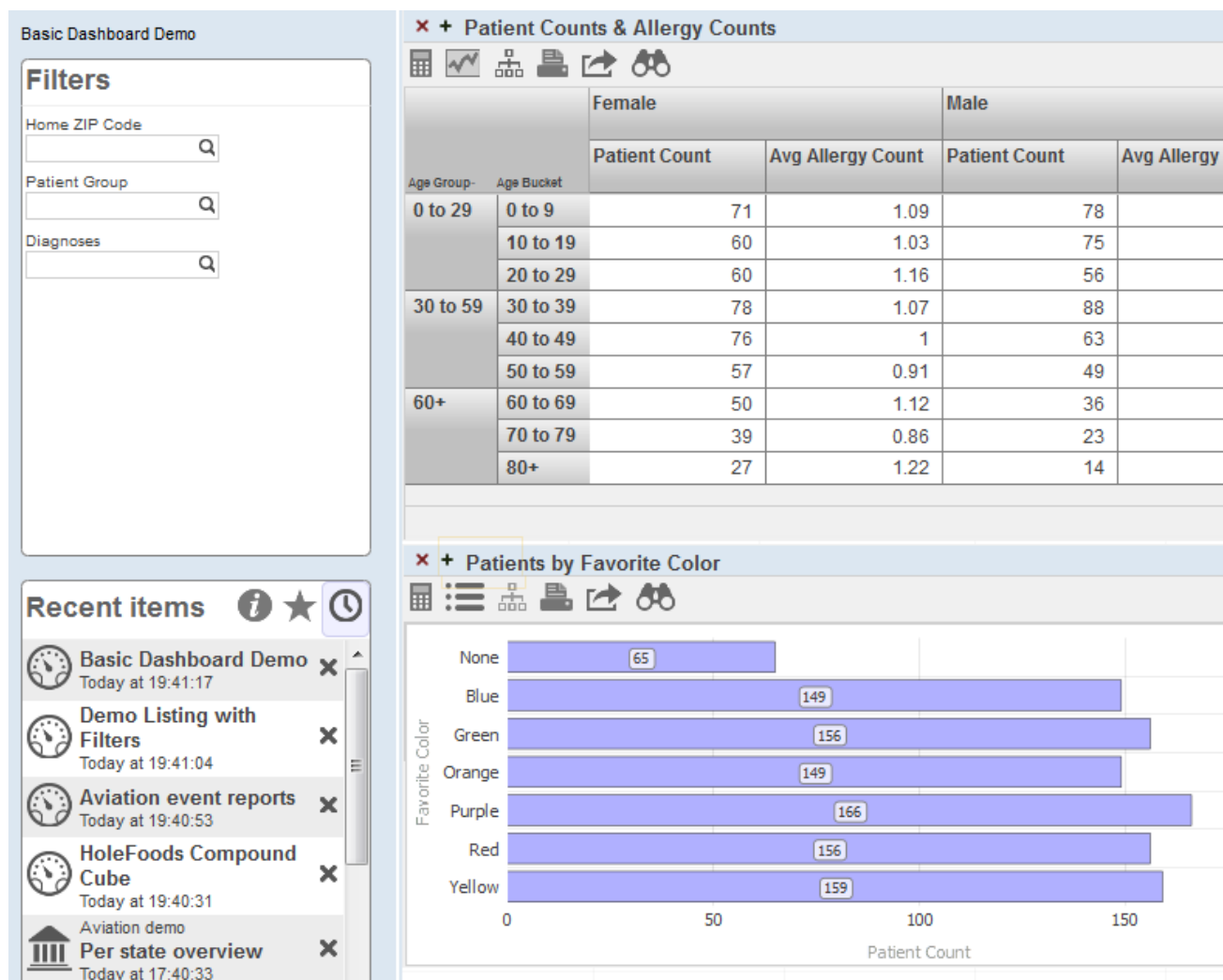
The purpose of InterSystems Business Intelligence is to enable you to embed business intelligence (BI) into your applications so that your users can ask and answer sophisticated questions of their data. Specifically, your application can include *dashboards*, which can include the following:

- Interactive widgets that execute queries designed for specific user roles or for specific areas of your user interface.
- Controls such as drop-down lists and data choosers that enable users to modify these queries.
- Interactive drill options that enable users to view the displayed data in different ways.
- Options to export, print, and send alerts to other users.
- An option to launch the Mini Analyzer, which supports free-form analysis.
- Execute custom actions that are provided as buttons or other controls.

In contrast to traditional BI systems that use static data warehouses, Business Intelligence is kept closely in synchronization with the live transactional data, as required for your business.

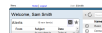
### 1.2 Dashboard Visual Details

The following example shows a sample dashboard:



A dashboard consists of the following areas:

- The upper left displays the name of the dashboard and (if defined) its title.
- Depending on the system configuration and on the individual layout of a dashboard, a dashboard can include zero, one, or two worklist areas on the left. For any worklist area, the upper right corner displays icons to indicate which worklists it can display. For example:



The highlighted icon indicates which worklist is currently displayed. You can select a different icon to display the corresponding worklist in this area instead.

The **Filters** worklist is specific to the dashboard. You use this to filter the widgets shown on this dashboard.

- The right area contains one or more widgets. Each *widget* is a rectangular panel that displays data in some form.

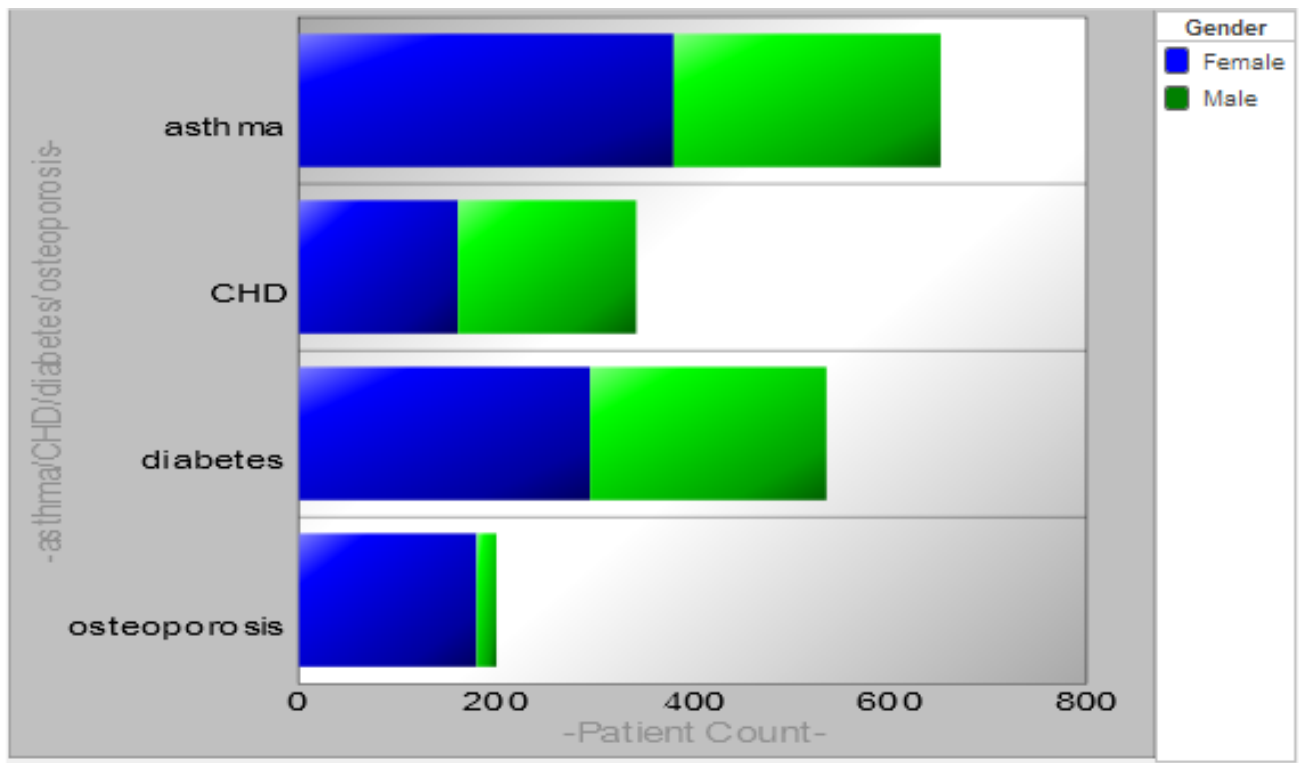
The following subsection describes the more common widgets.

## 1.2.1 Pivot Table Widgets

A pivot table widget displays data in one of three formats. First, it can display the data as a table:

| Age Group |          | Female        |                   | Male          |                   |
|-----------|----------|---------------|-------------------|---------------|-------------------|
|           |          | Patient Count | Avg Allergy Count | Patient Count | Avg Allergy Count |
| 0 to 29   | 0 to 9   | 680           | 0.60              | 750           | 0.63              |
|           | 10 to 19 | 756           | 0.66              | 769           | 0.69              |
|           | 20 to 29 | 661           | 0.64              | 648           | 0.61              |
| 30 to 59  | 30 to 39 | 815           | 0.63              | 735           | 0.65              |
|           | 40 to 49 | 728           | 0.68              | 741           | 0.61              |
|           | 50 to 59 | 586           | 0.58              | 552           | 0.62              |
| 60+       | 60 to 69 | 397           | 0.64              | 319           | 0.68              |
|           | 70 to 79 | 304           | 0.58              | 242           | 0.56              |
|           | 80+      | 217           | 0.57              | 100           | 0.66              |

Second, it can display the data as a chart:



Third, it can display a detail listing, which is a table that shows selected fields from the lowest-level records:

| <input type="checkbox"/> | #  | PatientID   | Age | Gender | Home City   | Test Score |
|--------------------------|----|-------------|-----|--------|-------------|------------|
| <input type="checkbox"/> | 1  | SUBJ_100631 | 0   | F      | Elm Heights | 50         |
| <input type="checkbox"/> | 2  | SUBJ_100781 | 0   | F      | Redwood     | 78         |
| <input type="checkbox"/> | 3  | SUBJ_100820 | 0   | F      | Magnolia    | 89         |
| <input type="checkbox"/> | 4  | SUBJ_100966 | 0   | F      | Cypress     | 91         |
| <input type="checkbox"/> | 5  | SUBJ_101274 | 0   | F      | Pine        |            |
| <input type="checkbox"/> | 6  | SUBJ_101340 | 0   | F      | Redwood     |            |
| <input type="checkbox"/> | 7  | SUBJ_101466 | 0   | F      | Magnolia    | 81         |
| <input type="checkbox"/> | 8  | SUBJ_101532 | 0   | F      | Pine        | 67         |
| <input type="checkbox"/> | 9  | SUBJ_101587 | 0   | F      | Elm Heights | 77         |
| <input type="checkbox"/> | 10 | SUBJ_102327 | 0   | F      | Redwood     | 79         |

## 1.2.2 Scorecard Widgets

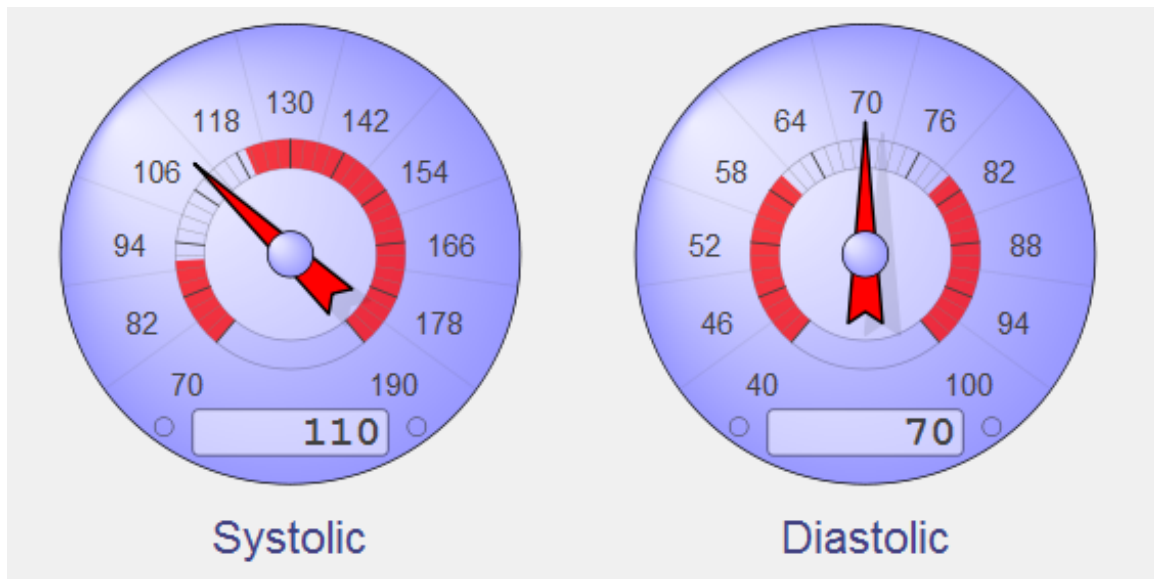
A scorecard widget displays one or more rows of data in a tabular format that also includes features such as value-dependent lamps and arrows. For example:



## 1.2.3 Meter Widgets

A meter widget displays one or more values, each in a graphical object as follows:





The preceding picture shows values in a speedometer. Business Intelligence supports several other forms of meters.

## 1.2.4 Map Widgets

A map widget shows a map with highlighted points that typically correspond to locations that are relevant to your business scenario:



## 1.2.5 Calendar Widgets

A dashboard can include an informational calendar widget like the following:

|          |     |      |       |     |     |     |
|----------|-----|------|-------|-----|-----|-----|
| «        | Day | Week | Month | »   |     |     |
| May 2011 |     |      |       |     |     |     |
| Sun      | Mon | Tue  | Wed   | Thu | Fri | Sat |
| 1        | 2   | 3    | 4     | 5   | 6   | 7   |
| 8        | 9   | 10   | 11    | 12  | 13  | 14  |
| 15       | 16  | 17   | 18    | 19  | 20  | 21  |
| 22       | 23  | 24   | 25    | 26  | 27  | 28  |
| 29       | 30  | 31   | 1     | 2   | 3   | 4   |

## 1.2.6 Custom Widgets

A dashboard can also include custom widgets called *portlets*. The following shows an example:

|           |      |     |
|-----------|------|-----|
| My Widget |      |     |
| Sales     | UP   | 12% |
| Costs     | DOWN | -8% |
| Profits   | UP   | 18% |

## 1.3 Data Sources for Widgets

In a dashboard, most widgets use a data source, which is one of the following:

- A pivot table. Pivot tables are created in the Analyzer. A pivot table is a query based on a cube, which is part of a Business Intelligence model. The [following section](#) discusses Business Intelligence models.
- A KPI (key performance indicator). A KPI is a more advanced query created by a programmer; it is also part of a Business Intelligence model.

## 1.4 Business Intelligence Models

A Business Intelligence model includes some or all of the following elements:

- At least one cube definition. A cube describes ways that you can query a set of specific base elements (such as patients or transactions). A cube includes *levels*, which enable you to group records from the base set, and *measures*, which show aggregate values of those records. It also defines listings and other items.

You use a cube to create pivot tables. For example:

| Patient Group | Avg Test Score |
|---------------|----------------|
| Group A       | 75.08          |
| Group B       | 74.22          |
| None          |                |

In this pivot table, the rows correspond to the members of the `Patient Group` level; each member is shown as one row. The data column displays the aggregate value of the `Avg Test Score` measure for each of these members; for this measure, the system computes the average value. Notice that the `Avg Test Score` is null for the `None` patient group.

- Any number of subject areas. A *subject area* is a subcube that enables users to focus on smaller sets of data without the need for multiple cubes. A subject area also enables you to customize captions and defaults of the cube.
- Any number of KPIs (key performance indicators). In Business Intelligence, a KPI is an interactive dataset that can be displayed on a dashboard. It uses a custom query created by a programmer. The query can use SQL, MDX (Multi-Dimensional Expressions, which is also generated by the Analyzer), or custom code.

The KPI can also define *actions*, which a user can launch and which execute your custom code.



# 2

## Introduction to the Business Intelligence User Interfaces

This page provides a quick look at the InterSystems IRIS Business Intelligence user interfaces.

**Note:** You can use Business Intelligence only in a specifically configured web application. See [Setting Up the Web Application](#). Also, be sure to consult *InterSystems Supported Platforms* for information on system requirements.

For information on the PMML Model Tester, see *Using PMML Models in InterSystems IRIS*.

### 2.1 Logging On to Business Intelligence

To log on to Business Intelligence, do the following in the Management Portal:

1. Switch to the appropriate namespace as follows:
  - a. Select the current namespace name to open the list of available namespaces.
  - b. Select the appropriate namespace from the list.
2. Select **Analytics**. The system displays a list of the tools for Business Intelligence:
  - **Architect** — Enables you to [define cubes](#).
  - **Analyzer** — Enables you to [define pivot tables](#).
  - **User Portal** — Launches the [User Portal](#), which includes the Analyzer and the Dashboard Designer.
  - **Tools** — Provides access to the [MDX Query Tool](#), the [Term List Manager](#), [Quality Measures](#), and the [Model Browser](#).
  - **Admin** — Provides access to the [Business Intelligence Logs](#), the [Folder Manager](#), [Settings](#), and the [Cube Manager](#).

**Note:** The Management Portal provides access to all Business Intelligence tools, including the back-end tools such as the Architect as well as the User Portal. Because the User Portal is intended for end users, it does not enable most users to return to the Management Portal.

## 2.2 Architect

The Architect enables you to define cubes and subject areas. You can use this tool or an IDE, or both together.

When you first display the Architect, using the sample HoleFoods cube, you see the following:

The screenshot shows the InterSystems Management Portal interface. At the top, the logo for InterSystems IRIS Data Platform is visible. The navigation bar includes links for Home, About, Help, Contact, and Logout. Below the navigation bar, the user is logged in as 'UnknownUser' and the instance is 'IRIS20201L'. The main area is titled 'Analytics > Architect - (HoleFoods)'. It features a 'Source Class' tree on the left with a list of fields like %ID, Actual, AmountOfSale, Channel, Comment, DateOfSale, Discount, Latitude, Longitude, Outlet, Product, TargetAmount, UnitsSold, and ZipCode. The central pane shows the 'Model Elements' for the 'HoleFoods' cube, categorized into Measures, Dimensions, and Comments. The Measures section includes 'Amount Sold', 'Units Sold', 'Max Units', 'Big Sale Count', and 'Comment'. The Dimensions section includes 'H1'. The right pane shows the 'Details' tab for the 'HoleFoods' cube, with fields for Cube Name, Display name, Description, Caption, Domain, Source class, Null replacement string, and Missing Value.

For details, see [Defining Models for InterSystems Business Intelligence](#).

For information on the permissions needed to use this tool, see [Setting Up Security](#).

## 2.3 Analyzer

The Analyzer enables you to define pivot tables. The Analyzer looks like the following.

The screenshot shows the InterSystems Management Portal interface for the Analyzer. The top navigation bar includes links for Home, About, Help, Contact, and Logout. The user is logged in as 'UnknownUser' and the instance is 'IRIS'. The main area is titled 'Analytics > Analyzer'. It features a 'View' section with icons for different views. The central pane shows a pivot table for 'HoleFoods Sales'. The 'Rows' section includes 'Product Category' and 'Units Sold'. The 'Columns' section includes 'Units Sold'. The 'Measures' section includes 'Units Sold'. The 'Filters' section is empty. The pivot table displays data for various product categories, including Candy, Cereal, Dairy, Fruit, Pasta, Seafood, Snack, and Vegetable, with their corresponding 'Units Sold' values.

For information, see [Using the Analyzer](#).

For information on the permissions needed to use this tool, see [Setting Up Security](#).

## 2.4 User Portal

The User Portal is intended for direct use by end users. The User Portal includes the Analyzer and the Dashboard Designer. The User Portal looks like the following:

The screenshot shows the User Portal interface for user Sam Smith. The top navigation bar includes a menu, home/logout links, the user name, and the license information. The main content area is divided into several sections:

- Welcome, Sam Smith**: A personalized greeting.
- Alerts**: A list of 19 new items. The table shows alerts from Keith Madison, Olivia Randolph-Erickson, and NBAKER. One alert from NBAKER is highlighted in red with the subject "PLEASE REVIEW THIS ASAP".
- Favorites**: A list of favorite dashboards, including "Basic Dashboard Demo" and "Sales Against Targets".
- Dashboard List**: A table of available dashboards with columns for Name, Type, and Keywords.

| Name                            | Type      | Keywords        |
|---------------------------------|-----------|-----------------|
| Basic Dashboard Demo            | DASHBOARD | Patients        |
| Demo Filter Interoperability    | DASHBOARD | Patients, KPIs  |
| Demo Linked Widgets             | DASHBOARD | HoleFoods       |
| Demo Listing with Filters       | DASHBOARD | Patients        |
| Demo OnClick Filter of Listing  | DASHBOARD | HoleFoods       |
| Demo Real Time Updates          | DASHBOARD | Patients, KPIs  |
| Demo Trend Lines                | DASHBOARD | Patients, KPIs  |
| Demo Two Subject Areas Together | DASHBOARD | Patients        |
| HoleFoods Compound Cube         | DASHBOARD | HoleFoods       |
| KPI with Crossjoin              | DASHBOARD | Patients, KPIs  |
| KPI with Listing                | DASHBOARD | HoleFoods       |
| KPI with Switchable Rows        | DASHBOARD | HoleFoods, KPIs |
| MDX Based KPI                   | DASHBOARD | Patients, KPIs  |
| Patients Compound Cube          | DASHBOARD | Patients        |
| Sales Against Targets           | DASHBOARD | HoleFoods, KPIs |
| Sample Bubble Chart             | DASHBOARD | Patients        |
| Sample Combo Chart              | DASHBOARD | Patients        |

For information, see [Using Dashboards and the User Portal](#).

For information on the permissions needed to use this tool, see [Setting Up Security](#).

## 2.5 MDX Query Tool

The Business Intelligence MDX Query Tool enables you to run ad hoc MDX queries. It looks like the following:

The screenshot shows the DeepSee Query Tool interface. At the top, there's a menu bar with 'Menu', 'Home | About | Help | Logout', and 'DeepSee > Query Tool'. Below this, a header bar displays 'HoleFoods' on the left, and 'Server: [redacted] Namespace: SAMPLES Switch' and 'User: SamSmith Licensed to: InterSystems Development Instance: CRML' on the right. A 'DeepSee' logo is also present.

The main area is divided into two sections. On the left, under 'View: Cube Members', there's a tree view with 'Measures' and 'Dimensions'. The 'Measures' list includes Count, Amount Sold, Units Sold, Max Units, Big Sale, Target, 5 year avg growth, Pct Sold, Percent of total, and Year over Year growth. The 'Dimensions' list includes Comments, DateOfSale, Product, Outlet, All Outlet, Region, and Country. On the right, the 'MDX statement' text box contains the query: `SELECT [Product].[P1].[Product Name].Members ON 1 FROM holefoods WHERE channel.retail`. Below the text box are three buttons: 'Execute', 'Show Plan', and 'Export to Excel'.

Below the buttons, a table displays the results of the query:

| Bagels (dozen)    | 29 |
|-------------------|----|
| Bundt Cake        | 28 |
| Calamari (frozen) | 8  |
| Cheerios (box)    | 31 |
| Donuts (dozen)    | 19 |

To execute an MDX query, type the query into the text box and then select **Execute**. You can also drag and drop items from the left area into the **MDX statement** area; if you do, the dropped items are added to the end of the query.

The bottom area on the right then displays the results.

To see the plan for the query, select **Show Plan**. For example:

The screenshot shows the 'Query execution plan' window. At the top, there are three buttons: 'Execute', 'Show Plan', and 'Export to Excel'. Below the buttons, the window displays the following information:

**Query execution plan:**

```
SELECT [%SEARCH] ON 0,[PRODUCT].[P1].[PRODUCT NAME].MEMBERS ON 1 FROM [HOLEFOODS].[RETAIL] WHERE [CHANNEL].[RETAIL]
```

**Execute dimension query: (%FindMemberByName)**

```
SELECT TOP 1 %ID,DxChannel MKEY,DxChannelExt FROM HoleFoods_Cube.StarChannel WHERE DxChannelExt=? ORDER BY DxChannelExt
```

**Execute dimension query: (%GetMembers)**

```
SELECT %ID,DxProduct MKEY, DxNameViaProduct FROM HoleFoods_Cube.StarProduct ORDER BY DxNameViaProduct
```

**Compute cell values in parallel**

1 (1 x 1) Tasks

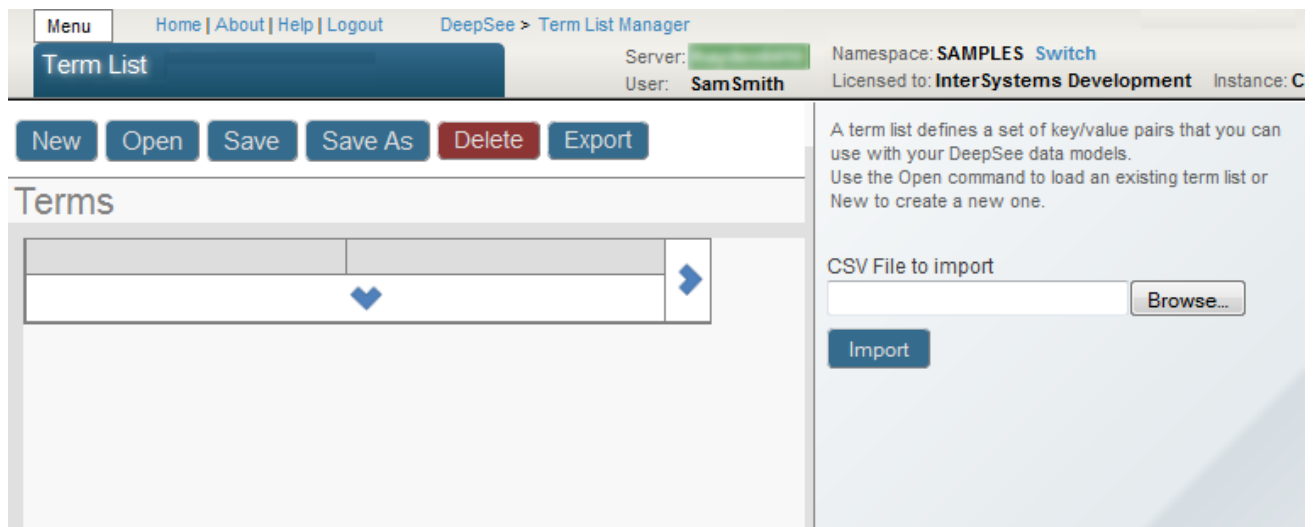
For an introduction to MDX, see [Using InterSystems MDX](#). For reference information on MDX, see the [InterSystems MDX Reference](#).

For information on the permissions needed to use this tool, see [Setting Up Security](#).

## 2.6 Term List Manager

The Term List Manager enables you to build term lists, which provide a way to modify a Business Intelligence model without programming. It looks like the following:





For information on creating term lists, see [Advanced Modeling for InterSystems Business Intelligence](#).

For information on the permissions needed to use this tool, see [Setting Up Security](#).

## 2.7 Listing Group Manager

The Listing Group Manager enables you to define listings that are not contained in any cube definition. The purpose of this tool is to enable you (and your customers, if appropriate) to define listings outside of cube definitions and without needing access to the Architect. The Listing Group Manager looks like this:

The screenshot shows a software interface for managing listing groups. At the top, there is a horizontal bar with six buttons: 'New', 'Open', 'Save', 'Save As', 'Compile', and 'Delete'. Below this, the interface is split into two main sections. The left section contains two buttons, 'Add Listing' and 'Remove Listing', followed by a tree view showing a hierarchy: 'Sample Listing Group' (expanded) containing 'Sample Listing 1' and 'Sample Listing 2'. The right section, titled 'Listing Group Details', contains several configuration fields: a 'Disabled' checkbox, a 'Listing Group Name' text field with the value 'Sample Listing Group', a 'Listing Group Display Name' text field, a 'Listing Group Class Name' text field with the value 'DeepSee.Model.Sample', a 'Listing Group Target Cubes' text field with the value 'PATIENTS,RELATEDCUI' and a search icon, a 'Listing Group Resource' dropdown menu, a 'Default Resource For Listings' dropdown menu, and a 'Group Description' text area with a file icon.

For information, see [Defining Listing Groups](#).

For information on the permissions needed to use this tool, see [Setting Up Security](#).

## 2.8 Quality Measure Manager

The Quality Measure Manager enables you to define quality measures, a kind of calculated measure that can be reused in multiple contexts. It looks like the following:

Menu

Home | About | Help | Logout

Analytics > Quality Measure Manager

Quality Measures

Server: **SAMPLES**  
User: **SamSmith**

Namespace: **SAMPLES** [Switch](#)  
Licensed to: **InterSystems Development**

Browse Open Edit New Save Save As Remove

The complete set of available metrics are displayed on the right.

Catalog

Preventive health care

Children and adolescents

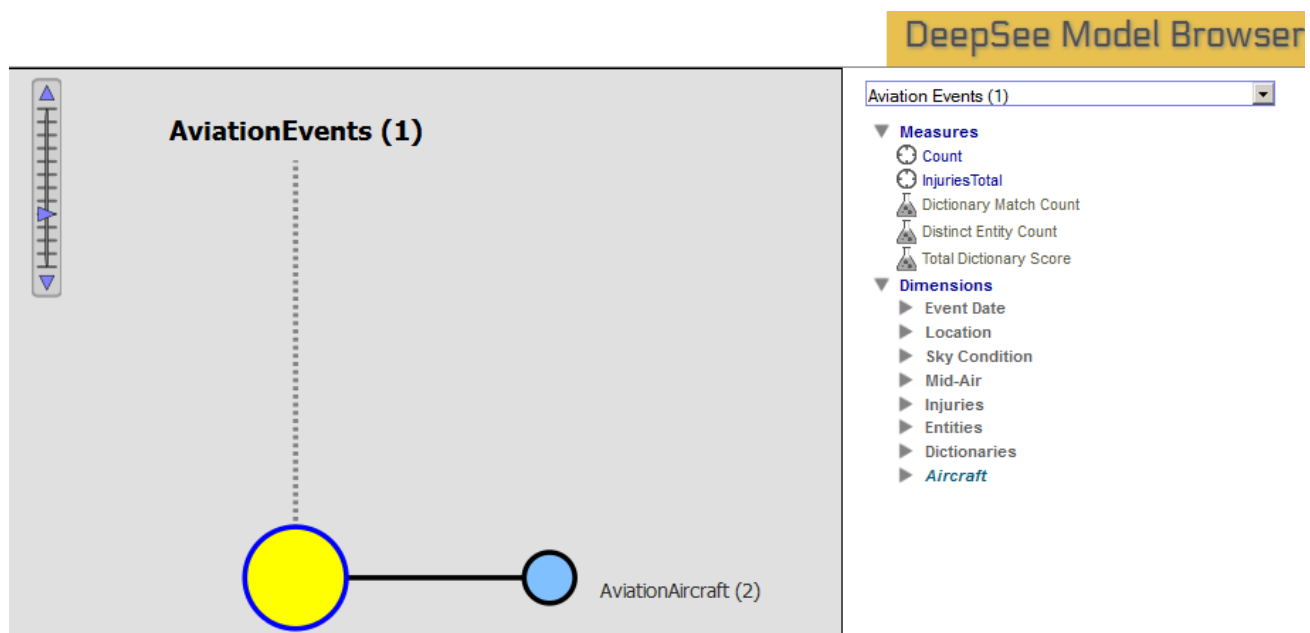
|                                                                                                                                        |                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>WA</b><br>Weight assessment and counseling for nutrition and physical activity for children and adolescents<br><a href="#">View</a> | <b>ADHD</b><br>Follow-up care for children prescribed ADHD medication<br><a href="#">View</a> |
| <b>WC0-15m</b><br>Well-child exams (0-15 months)<br><a href="#">View</a>                                                               | <b>WC3-6y</b><br>Well-child exams (3-6 years)<br><a href="#">View</a>                         |
| <b>WC12-21y</b><br>Well-child exams (12-21 years)<br><a href="#">View</a>                                                              | <b>IMM-CH</b><br>Childhood immunization<br><a href="#">View</a>                               |
| <b>IMM-AD</b><br>Immunizations for Adolescents<br><a href="#">View</a>                                                                 | <b>LEADSCR</b><br>Lead screening in children (Medicaid only)<br><a href="#">View</a>          |

For information, see [Advanced Modeling for InterSystems Business Intelligence](#)

For information on the permissions needed to use this tool, see [Setting Up Security](#).

## 2.9 Model Browser

The Model Browser is a useful way of viewing relationships among cubes. It looks like the following:



For information, see [Defining Models for InterSystems Business Intelligence](#).

For information on the permissions needed to use this tool, see [Setting Up Security](#).

## 2.10 Business Intelligence Logs

The **Logs** option displays the Business Intelligence log file, which the system generates when it builds cubes. It looks like the following:

The screenshot shows the Log File Viewer interface. At the top, there is a navigation bar with 'Menu', 'Home | About | Help | Logout', and 'DeepSee > Log File Viewer'. Below this, a 'Log File Viewer' section shows the 'Server' as 'UnknownUser' and 'Namespace' as 'SAMPLES'. There are 'Refresh' and 'Delete' buttons. The main content area displays the log file path 'C:\InterSystems\Cache\mgr\DeepSeeTasks\_SAMPLES.log' and a list of log entries:

```

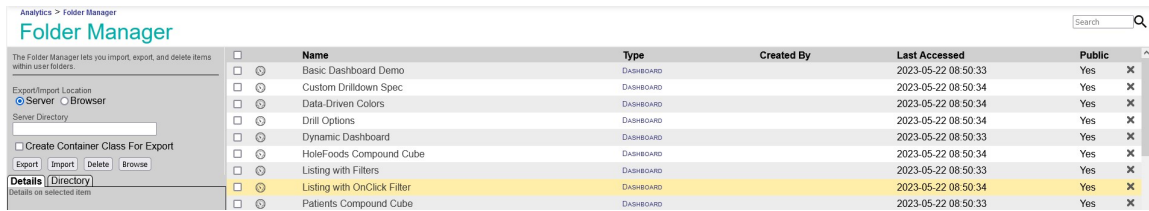
2013-06-11 15:33:35
-----
2013-02-22 17:47:35.067      8548 TaskMaster Create background agents..
2013-02-22 17:47:35.267      8548 TaskMaster Background agents created: 16
2013-03-01 09:12:49.677      6932 TaskMaster Create background agents..
2013-03-01 09:12:49.924      6932 TaskMaster Background agents created: 16

```

For information on the permissions needed to access this page, see [Setting Up Security](#).

## 2.11 Folder Manager

The Folder Manager enables you to manage items within user folders. It looks like the following:



You can use this to export pivot tables and dashboards so that you can package their definitions into a class definition. See [Implementing InterSystems Business Intelligence](#).

For information on the permissions needed to use this tool, see [Setting Up Security](#).

## 2.12 Settings

The **Settings** option lets you specify settings that affect the appearance of Business Intelligence within this namespace. It looks like the following:

Menu

Home | About | Help | Logout

Analytics > Settings

Settings

Server: **SAMPLES**    Namespace: **SA**  
User: **Sam Smith**    Licensed to: **Int**

Save

### User Portal settings for namespace SAMPLES.

Press Save to apply changes.

General

Worklists

Run-time Variables

User-defined Icons

General Color Scheme  
Simple

Chart Series Color Scheme  
Default

Home page title

Title for Portal Home page

Company Name

Company name to display in Portal title.

Company Logo

URL of icon to display in Portal title.

For information, see [Implementing InterSystems Business Intelligence](#).

For information on the permissions needed to use this tool, see [Setting Up Security](#).

## 2.13 Cube Manager

The Cube Manager enables you to easily update cubes. Using the Cube Manager, you can define a schedule for when you want to update your cubes. The Cube Manager then adds automated tasks that rebuild and (when supported) synchronize your cubes at the dates and times which are specified by the schedule. It looks like the following:

**Cube Manager** Expand All Collapse All View:  

Filter

- ▼ Cubes
  - AVIATIONAIRCRAFT
  - AVIATIONCREW
  - AVIATIONEVENTS
  - CITIES
  - CITYRAINFALL
  - COMPOUNDCUBE/CITYRAINFALL
  - COMPOUNDCUBE/DOCTORS
  - COMPOUNDCUBE/PATIENTS
  - CONNECTORCUBE
  - HOLEFOODS
  - HOLEFOODSBUDGET
  - PATIENTS
  - PATIENTSQUERYCUBE
  - RELATEDCUBES/ALLERGIES
  - RELATEDCUBES/CITIES
  - RELATEDCUBES/CITYRAINFALL
  - RELATEDCUBES/DOCTORS
  - RELATEDCUBES/PATIENTS

| Scheduled Cubes           | Status  | Date/Time                   | Time To Complete |
|---------------------------|---------|-----------------------------|------------------|
| ▼ Weekly Doctor Update    |         | Next: 6/22/2025, 2:00:00 AM | 25.20 s          |
| RELATEDCUBES/CITIES       | Success | 6/5/2025, 3:44:54 PM        | less than 1 s    |
| RELATEDCUBES/DOCTORS      | Success | 6/16/2025, 7:09:27 AM       | 10.52 s          |
| ▼ Nightly Patient Update  |         | Next: 6/18/2025, 3:00:00 AM | less than 1 s    |
| COMPOUNDCUBE/PATIENTS     | Success | 6/17/2025, 12:59:46 AM      | less than 1 s    |
| COMPOUNDCUBE/COMPOUNDCUBE |         |                             |                  |

Details Tools

Registered Group

Name

Weekly Doctor Update

Disabled ☐

Update Plan

☐ Every  
☐ Hourly At  
☐ Daily At  
☒ Weekly On  
☐ Monthly On

Week Day

Sunday at 02:00 AM

Maintenance

Build This Group

| Cube                      | Size   | Time To Com   |
|---------------------------|--------|---------------|
| RELATEDCUBES/ALLERGIES    | 753    | 3.            |
| RELATEDCUBES/CITIES       | 9      | less than 1 s |
| RELATEDCUBES/CITYRAINFALL | 13,608 | 10.           |
| RELATEDCUBES/DOCTORS      | 40     | 10.           |

For details, see [Using the Cube Manager](#).

For information on the permissions needed to use this tool, see [Setting Up Security](#).

# 3

## Introduction to the Other Business Intelligence Tools

This page introduces the other tools for working with InterSystems IRIS Business Intelligence.

### 3.1 BI Samples

Most of the samples in this documentation are part of the Samples-BI sample (<https://github.com/interSystems/Samples-BI>) or the Samples-Aviation sample (<https://github.com/interSystems/Samples-Aviation>).

InterSystems recommends that you create a dedicated namespace called SAMPLES (for example) and load samples into that namespace. For the general process, see *Downloading Samples for Use with InterSystems IRIS*.

### 3.2 MDX Shell

The system provides a shell in which you can issue MDX queries to explore your cubes and subject areas. This section introduces this shell and lists the supported MDX options and functions.

For an introduction to MDX queries, see *Using InterSystems MDX*, which contains many examples.

Also see the *InterSystems MDX Reference*.

#### 3.2.1 Accessing the MDX Shell

To access the MDX shell, start the Terminal and do the following:

1. Switch to the namespace in which you defined the cube or subject area.
2. Enter the following command:

##### **ObjectScript**

```
Do ##class(%DeepSee.Utils).%Shell()
```

Now you can enter MDX queries like the following:

```
SELECT MEASURES.[%COUNT] ON 0, birthd.decade.MEMBERS ON 1 FROM patients
```

When you do so, the shell executes the query, displays its results to the console, and redisplay the shell prompt, as follows:

```

                                Patient Count
1 1910s                                71
2 1920s                                223
3 1930s                                572
4 1940s                                683
5 1950s                               1,030
6 1960s                               1,500
7 1970s                               1,520
8 1980s                               1,400
9 1990s                               1,413
10 2000s                              1,433
11 2010s                               155
-----
Elapsed time:                .014128s
```

In the shell:

- To display a list of cubes and subject areas, enter `cube`
- To see the contents of a cube or subject area, enter `cube name_of_cube_or_subject_area`

**Note:** This command does not display calculated members and named sets, although you can use these elements in the shell and elsewhere.

For a subject area, this command lists all elements, even if those are specified as hidden in the subject area.

- To exit the shell, enter `q`
- To enable query caching, enter `cache on`. If you set `cache off` and then run an MDX query in the shell, the MDX shell purges all cached queries for that specific cube system-wide. This could lead to slower performance for other users.
- To enable the asynchronous mode, enter `async on`
- To build a cube, enter `build cubename`
- To reset the query cache, enter `reset`
- For a list of additional shell options, enter `?`

## 3.2.2 Viewing the Indexes Used by a Query

The Business Intelligence shell provides a quick way to see the indexes that a query uses:

1. Issue the following shell command:

```
stats on
```

2. Enter the query, preceded by `%SHOWPLAN`. For example:

```
%SHOWPLAN SELECT aged.[age group].members ON 0, allerd.H1.MEMBERS ON 1 FROM patients WHERE colord.red
```

```

                                0 to 29          30 to 59          60 +
1 additive/colorin                27             19             14
2 animal dander                   15             25              8
3 ant bites                       15             19             11
4 bee stings                      24             27              7
5 dairy products                  25             25              4
6 dust mites                      28             23             10
7 eggs                           19             21             13
8 fish                           26             17             11
```



```

9 mold 23 23 6
10 nil known allerg 80 82 21
11 No Data Availabl 216 194 92
12 peanuts 26 15 8
13 pollen 29 22 11
14 shellfish 29 23 14
15 soy 25 25 6
16 tree nuts 22 18 8
17 wheat 16 17 8
----- Query Plan -----
**%SHOWPLAN SELECT [AGED].[AGE GROUP].MEMBERS ON 0,[ALLERD].[H1].MEMBERS ON 1 FROM [PATIENTS] WHERE
[COLORD].[RED]**
**DIMENSION QUERY (%FindMemberByName): SELECT TOP 1 %ID,Dx327553094 MKEY,Dx327553094 FROM
Cubes_StudyPatients.Star327553094 WHERE Dx327553094=? ORDER BY Dx327553094**
**EXECUTE PARALLEL: 1x1 task(s) **
**CONSOLIDATE**
----- End of Plan -----

```

Line breaks were added here for readability.

The system captures all the indexes used by the query and reports them. Note that the query results are not necessarily correct because the query is only partially run; the purpose of %SHOWPLAN is to enable you to see the indexes, not to get the query results.

## 3.3 Utility Methods

- The class %SYSTEM.DeepSee includes the most commonly used utility methods. These include:

- **BuildCube()**
- **KillCube()**
- **ListCubes()**
- **Reset()**
- **Shell()**
- **SynchronizeCube()**

This class is available via the special variable **\$SYSTEM**, as are all classes in the %SYSTEM package. For example, to build a cube, you can use the following:

### ObjectScript

```
Do $system.DeepSee.BuildCube("MyCube")
```

- The class %DeepSee.Utils includes a large set of utility methods, including:
  - **%ExportExcelToFile()** — exports a Business Intelligence query or KPI to a file in Microsoft Excel format
  - **%ExportPDFToFile()** — exports a Business Intelligence query or KPI to a file in PDF format
  - **%GetAgentCount()** — gets the current agent count
  - **%GetBaseCube()** — gets the name of cube on which a subject area is based
  - **%GetCubeFactClass()** — gets the name of fact table class associated with a cube
  - **%GetCubeLevels()** — gets the levels, measures, and relationships defined in a cube
  - **%GetDimensionMembers()** — gets the list of members of a dimension
  - **%GetMetricList()** — gets all production business metrics visible to current user

- **%GetSQLTableName()** — gets SQL table name for a given class
  - **%ProcessFact()** — updates a single fact for a cube
  - **%GetMDXFromPivot()** — returns the MDX query defined by a pivot table
  - **%ExecutePivot()** — runs the MDX query defined by a pivot table and optionally returns an instance of `%DeepSee.ResultSet`
  - **%GetResultsetFromPivot()** — returns an instance of `%DeepSee.ResultSet` that holds the MDX query defined by a pivot table and optionally runs that query
- The class `%DeepSee.UserLibrary.Utils` includes methods that you can use to programmatically perform the tasks supported in the Folder Manager. These methods include:
    - **%AddFavorite()**
    - **%DeleteFolderContents()**
    - **%DeleteFolderItem()**
    - **%Export()**
    - **%GetFolderList()**
    - **%ImportContainer()**

## 3.4 Data Connector

The data connector class (`%DeepSee.DataConnector`) enables you to make arbitrary SQL queries available for use in cubes and listings. See [Implementing InterSystems Business Intelligence](#).

## 3.5 Result Set API

The class `%DeepSee.ResultSet` enables you to execute MDX queries programmatically and access the results.

For information, see [Implementing InterSystems Business Intelligence](#).

## 3.6 JavaScript and REST APIs

The Business Intelligence JavaScript API is provided by the file `DeepSee.js`, which is in the `install-dir/CSP/broker` directory. This JavaScript library enables you to interact with Business Intelligence from a client that is based on JavaScript. The functions in this library are a wrapper for a REST-based API for Business Intelligence. You can also use the REST API directly.

For information, see [Client-Side APIs for InterSystems Business Intelligence](#).

# Business Intelligence Glossary

This glossary summarizes terms found in the InterSystems IRIS Business Intelligence documentation. If you have not yet done so, InterSystems highly recommends that you read [Basic Concepts](#).

## action

An operation that a user can start by using a [control](#) (such as a button) on a dashboard. The system provides a set of standard actions (such as applying a filter, navigating to another dashboard, and others), and you can add custom actions. See [Defining Custom Actions](#).

## age dimension and age level

An age dimension is a [dimension](#) that contains age [levels](#). An age level groups data by an age, relative to the cube build time, computed from a date or time value in the source data. Age dimensions and age levels are not generally recommended, because they require nightly [rebuids](#).

## age measure

A [measure](#) that provides an aggregated age value in days. Age measures are not generally recommended, because they require nightly [rebuids](#).

## All level and All member

The All level is a special, optional [level](#), which appears in all the hierarchies of a [dimension](#). If defined, this level contains one [member](#), the All member, which corresponds to all records in the cube. You can use the All member to create a summary line in a pivot table.

## BI

Business intelligence, a set of tools and techniques that transform raw data into insights that can improve the operation of a business or other organization. BI is intended to support a measurement-based approach to making strategic and tactical decisions.

## building a cube

The process of iterating through the [source class](#) for a cube and populating the [fact table](#) (and building the indexes for that table). See also [synchronizing a cube](#).

For details, see [Compiling and Building Cubes](#) and [Keeping the Cubes Current](#).

## business metric

A two-dimensional array of data generated by a running production and generally providing data relevant to or about that production. Like [pivot tables](#), business metrics can be displayed on a [dashboard](#), within a [widget](#). For information on creating production business metrics, see *Developing Productions*.

## business rule

A concept that allows nontechnical users to change the behavior of business processes within a production. You can use them in source expressions in cubes; see [Details for Source Expressions](#). For details on production business rules, see *Developing Business Rules*.

**calculated measure**

A measure that is based on other measures via an [MDX](#) expression. The phrase *calculated measure* is not standard in MDX, but this documentation uses it for brevity. Formally, a calculated measure is a [calculated member](#) that belongs to the Measures dimension.

**calculated member**

A member that is based on other members via an [MDX](#) expression. You can define two kinds of calculated members:

- A calculated measure is a measure is based on other measures. (In MDX, each measure is a member of the Measures dimension.)

For example, one measure might be defined as a second measure divided by a third measure.

The phrase *calculated measure* is not standard in MDX, but this documentation uses it for brevity.

- A non-measure calculated member typically aggregates together other non-measure members. Like other non-measure members, this calculated member is a group of records in the fact table.

See [Defining Calculated Members](#).

**compound cube**

A special kind of [subject area](#) that combines multiple cube definitions (typically two) and that enables you to create pivot tables that contain elements from multiple cubes. See [Defining Shared Dimensions and Compound Cubes](#).

**computed dimension**

A special kind of Business Intelligence dimension whose members are computed at runtime via an SQL or [MDX](#) expression. See [Defining Computed Dimensions](#).

Computed dimensions do not have any association with calculated members. A computed dimension is specific to Business Intelligence. A calculated member is a standard concept in MDX.

**container class**

A class that extends %DeepSee.UserLibrary.Container. This class can contain the definitions of pivot tables, dashboards, and other Business Intelligence [folder items](#). When you compile this class, the system generates those folder items, replacing any current definitions that they might have. See [Implementing InterSystems Business Intelligence](#).

**control**

An interactive element on a [dashboard](#). Controls include drop-down lists and buttons.

**cube**

An model of your data that defines elements that can be used in [MDX](#) queries. These elements determine how you can query your data, specifically, a set of specific records (such as patient records or transaction records). The set of records is determined by the [source class](#) for the cube. For an introduction, see [Basic Concepts](#).

**cube inheritance**

A mechanism in Business Intelligence that enables you to define multiple similar cubes. This mechanism has no relationship to class inheritance. See [Using Cube Inheritance](#).

## custom listing

A [listing](#), specifically one of the following special kinds of listings:

- A listing that uses a custom SQL query that retrieves fields from some other table, not the source table used by the cube, and not a data connector. See [Defining Listings](#).
- A listing that consists of listing fields chosen by the user, in the Analyzer. See [Performing Ad Hoc Analysis](#).

## dashboard

An interactive display of data, particularly data that provides a high-level data of a business. See [Creating Dashboards](#).

## data connector

A class that extends %DeepSee.DataConnector. A data connector maps the results of an arbitrary SQL query into an object that can be used as the source of a [cube](#). Typically, a data connector accesses external data not in an InterSystems database, but you can also use it to specify an SQL query against an InterSystems database, including an SQL query on a view. See [Defining and Using Data Connectors](#).

## detail listing

See [listing](#).

## dimension

A container for [levels](#). A dimension contains one or more [hierarchies](#), which in turn contain levels. For example, a single dimension might contain multiple hierarchies related to allergies. There is no formal relationship between two different hierarchies or between the levels of one hierarchy and the levels of another hierarchy. The practical purpose of a dimension is to define the default behavior of the levels that it contains — specifically the All level.

See [Defining Dimensions, Hierarchies, and Levels](#).

## dimension table

The table in which Business Intelligence stores the members of a level and any properties they have. See [Details for the Fact and Dimension Tables](#).

## drill down

Examine a row of a pivot table and see the data for that row displayed in a more granular way. For example, a row might display data for a year, and you would drill down to see data for that year, broken out by month. Business Intelligence supports multiple forms of drill down. See [Performing Ad Hoc Analysis](#).

Informally (although not in this documentation), the phrases drill down and [drill through](#) are sometimes used interchangeably, and it is wise to double-check which phrase is intended.

## drill through

Formally, *to drill through* means to display a [listing](#). Internally, the system uses the MDX [DRILLTHROUGH](#) statement when it displays a listing. See [Performing Ad Hoc Analysis](#).

Informally (although not in this documentation), the phrases drill through and [drill down](#) are sometimes used interchangeably, and it is wise to double-check which phrase is intended.

**expression**

An expression (<expression> element) whose value is available while the system is [building](#) a row in the [fact table](#). You can define an expression that uses complex or time-consuming logic, and then you can base multiple cube elements on the expression. Expressions are for use during cube build only and are provided for efficiency.

See [Other Options](#).

**fact**

A row in the [fact table](#).

**fact table**

A generated structure that the system queries directly. When you compile a [cube](#) definition, the system generates a fact table class. When you build a cube, the system creates records for this table and indexes them. See [Basic Concepts](#).

**filter**

A restriction on the data. The system provides two simple ways to filter data: member-based filters and measure-based filters. You can combine these, and more complex filters are also possible, especially if you write MDX queries directly. For an introduction, see [Filters](#).

**folder item**

Any of the following Business Intelligence items:

- Pivot tables
- Saved widgets
- Dashboards
- Themes

Business Intelligence folder items are accessible within [your IDE](#) as generated .dfi files. If you are using [Visual Studio Code](#), you can access these pivot table definitions in the following ways:

- Client-side editing: when a namespace is opened in the [ObjectScript Explorer](#), view the .dfi files in the Other directory.
- Server-side editing: modify your [workspace filters](#) so that they show generated files and include the desired .dfi files.

**geo listing**

See [map listing](#).

**hierarchy**

An organization of [levels](#). Levels belong to hierarchies (which belong to [dimensions](#)). A hierarchy can contain only single level or can contain multiple levels. If it contains multiple levels, the higher levels of the hierarchy are less granular than the lower levels. That is, each [member](#) of a higher level contains a larger set of records than does a member of a lower level.

In casual usage, a higher level is called the parent of the lower level. However, it is useful to remember that the hierarchy is actually a hierarchy among members. Thus it is more accurate to state that a member of the higher level is the parent of one or more members of the lower level. Conversely, any member of a lower level is the child of exactly one member of the higher level.

Hierarchies provide additional features beyond those provided by levels; see [Hierarchies and Dimensions](#) and [Defining Dimensions, Hierarchies, and Levels](#).

## KPI

A class based on %DeepSee.KPI. In most cases, a KPI uses a query and displays a result set. Like [pivot tables](#), KPIs can be displayed on a [dashboard](#), within a [widget](#). You can also use KPIs as building blocks for [calculated members](#) (including calculated measures). Start with [Defining Basic KPIs](#).

## level

A [cube](#) element that enables you to group records. A level consists of members, each of which is a set of records. See [Basic Concepts](#) and [Details of Defining Levels](#).

## list-based level

A [level](#) that is based upon a list value. For example, a patient can have multiple diagnoses. The `Diagnoses` level groups patients by diagnosis. With a list level, it is possible for a given record of the source class to have multiple values and thus for that record to belong to multiple members of the level.

## listing

An SQL query that accesses the lowest-level records associated with one or more cells of a pivot table. See [Defining Listings](#).

## listing field

A `<listingField>` element defined in a cube definition. Users can select the listing fields to include, when they create [custom listings](#) in the Analyzer. See [Defining Listing Fields](#).

This phrase can also refer more generally to any field in any listing.

## listing group

A class that defines a group of listings. Listing groups are created in the Listing Group Manager. The purpose of this tool is to enable you (and your customers, if appropriate) to define listings outside of cube definitions and without needing access to the Architect. See [Defining Listing Groups](#).

## map listing

A [listing](#) that contains location data and is displayed as a map. Each pin on the map corresponds to a source record.

**Important:** A map listing uses the Google Maps API. Be sure that your usage of this API is consistent with the Terms of Use, which you can access via a link displayed in this listing.

Note that in order to use the Google Maps API, you must obtain an API key. See [Specifying Basic Settings](#).

## map widget

A [dashboard widget](#) that contains location data and is displayed as a map. Each pin on the map corresponds to a member of a level, particularly a level that refers to locations.

**Important:** A map widget uses the Google Maps API. Be sure that your usage of this API is consistent with the Terms of Use, which you can access via a link displayed in this widget.

Note that in order to use the Google Maps API, you must obtain an API key. See [Specifying Basic Settings](#).

**MDX**

MultiDimensional eXpressions, a standard query language for OLAP (online analytical processing) databases and used in many [BI](#) applications. See [Using InterSystems MDX](#) and [InterSystems MDX Reference](#).

**measure**

A [cube](#) element that (with rare exceptions) aggregates values across multiple records. Each measure is based on a source value, which is either a class property or an ObjectScript expression. The definition of a measure also includes an aggregation function, which specifies how to aggregate values for this measure. See [Basic Concepts](#) and [Defining Measures](#).

**member**

A set of records. Every [level](#) has one or members. See [Basic Concepts](#) and [Details of Defining Levels](#).

**named filter**

A reusable [filter](#) that is defined in the Analyzer. See [Filtering Pivot Tables](#).

**named set**

A reusable MDX [set](#) that is defined within a [cube](#). See [Defining Named Sets](#).

**NLP dimension**

A special kind of [dimension](#) that analyzes an [NLP measure](#), which in turn is a measure based on [unstructured text](#). See [Using Text Analytics in Cubes](#).

**NLP measure**

A special kind of [measure](#) that is based on [unstructured text](#). You cannot display NLP measures directly in pivot tables. Their purpose is to provide data for use by [NLP dimensions](#). See [Using Text Analytics in Cubes](#).

**pivot table**

An interactive, drillable display of data, generally with rows and columns, designed for specific user roles or for specific areas of your user interface. A pivot table is based on an MDX query that is executed at runtime can respond to input such as filter selections made by the user. Internally it obtains values from a [cube](#). See [Using the Analyzer](#).

**pivot variable**

An element that is intended to be used in pivot tables, specifically, in selected parts of the query that defines the pivot table. When a dashboard displays the pivot table, that dashboard can include a [control](#) with which the user can change the value of the corresponding pivot variable. See [Defining and Using Pivot Variables](#).

Pivot variables are entirely different from [runtime variables](#).

**portlet**

A custom [widget](#) that can be displayed on [dashboards](#). For information on creating portlets, see [Implementing InterSystems Business Intelligence](#).



## plugin

A specialized form of [KPI](#) that defines one or more computations to use in the Analyzer and in queries. Plug-ins are especially appropriate for complex or time-consuming computations. For example, you might have a computation that uses several different parts of the source record, as well as external information; a plug-in would be suitable in this case. See [Defining Plug-ins](#).

## property

A value that is specific to a [member](#) of a given [level](#). If a level has a property, then each member of that level has a value for that property; other levels do not have values for the property. You can use properties in queries in much the same way that you use measures. In Business Intelligence, you can also use properties for other purposes such as controlling member names and controlling the order in which member are sorted. See [Defining Properties](#).

## quality measure

A quality measure is similar to a [calculated measure](#) because it is defined by a formula that combines [MDX](#) expressions. You specify the subject area or subject areas in which it is available, and you can control whether the quality measure is *published* (and thus available in the Analyzer). Each quality measure is a subclass of %DeepSee.QualityMeasure.QualityMeasure.

For information, see [Defining Quality Measures](#).

## related cube

A [cube](#) whose [dimensions](#), [hierarchies](#), and [levels](#) are available within another cube, because there is a [relationship](#) between the two cubes.

## relationship

A connection between two cubes that makes the dimensions of one cube available in the other cube (and possibly vice versa). If you define relationships, you can define a level once rather than multiple times, which minimizes the sizes of fact tables and their indexes. See [Cube-Cube Relationships](#).

## runtime variable

A special element that is intended for use as the default value of a filter on a dashboard (currently this is their only use). The definition of a runtime variable is an ObjectScript expression that is evaluated at runtime. See [Configuring Settings](#).

Runtime variables are entirely different from [pivot variables](#).

## searchable measure

A [measure](#) that enables you to apply a [filter](#) that considers the values in the [source records](#). Searchable measures are an InterSystems extension to MDX. In standard MDX, a filter can be based only on [members](#). See [Defining Measures](#).

## Selective Build

A cube build option that allows you to build a specific level, measure, or relationship in a cube, without having to rebuild the entire cube. See [Using Selective Build](#).

## set

A list of multiple MDX items, typically used for rows or columns of a pivot table. The items can be any combination of literal values, [members](#), and [tuples](#). For an introduction, see [Working with Sets](#). For reference information, see [Set Expressions](#).

**shared dimension**

A dimension that can be used in more than one cube. That is, more than one cube can use members of the dimension for rows or columns or for filtering. A dimension can be shared formally or informally. If the dimension is shared formally, you can define a [compound cube](#) that combines the cubes that use this dimension. See [Defining Shared Dimensions and Compound Cubes](#).

**source class, source records**

The source class is the class that contains the data upon which a [cube](#) is based. Every cube has a source class, which is usually a persistent class. A source class has a set of source records. For an introduction, see [Basic Concepts](#).

**star table**

See [dimension table](#).

**subject area**

A view of a [cube](#) with optional overrides. A subject area uses the fact table and related tables of the associated cube and does not require independent updates. You define subject areas to enable users to focus on smaller sets of data without the need for multiple cubes. See [Defining Subject Areas](#).

**synchronizing a cube**

The process of updating the [fact table](#) and indexes for a [cube](#), based on incremental changes to the source class. See [Compiling and Building Cubes](#) and [Keeping the Cubes Current](#).

See also [building a cube](#).

**term list**

A simple (but extendable) list of key and value pairs. Term lists provide a way to customize a Business Intelligence model without programming. See [Defining Term Lists](#).

**time dimension and time level**

A time dimension is a [dimension](#) that contains time [levels](#). A time level groups data by a date or time value in the source data.

**tuple**

A type of MDX value that consists of an intersection of members. If the tuple refers to each dimension in the cube, the tuple is *fully qualified*. Otherwise, it is *partially qualified*.

For an introduction, see [Tuples and Cubes](#). For reference information, see [Tuple Expressions](#).

**unstructured data**

Data that is written as text in a human language such as English or French. The Analytics Engine analyzes unstructured data. For a general introduction, see Conceptual Overview.

You can use unstructured data within cubes, if the source table for a cube includes a property that contains unstructured data. See [Using Text Analytics in Cubes](#).

**widget**

A rectangular area that lies within a [dashboard](#) and that (in most cases) displays data obtained from Business Intelligence. See [Creating Dashboards](#).